PHD125 Pharmaceutical Organic Chemistry · Carbohydrates

Tutorial: exam-style practice

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

Part A: 12 multiple-choice questions (1 mark each). Part B: 5 structured questions (38 marks). Same format as the final exam.

Answer Part A by circling one option. Answer Part B in full, showing your working. The answer key is at the back.

Part A · Multiple choice

Choose one answer for each question.

A1 Topic: Carbohydrates CH5-LO1

Carbohydrates are best defined as:

A. polyhydroxy aldehydes or ketones, or compounds that hydrolyse to them
B. compounds of carbon and water molecules
C. esters of glycerol
D. polymers of amino acids

A2 Topic: Carbohydrates CH5-LO1

Enantiomers differ in their:

A. direction of rotation of plane-polarised light
B. boiling points
C. molecular formulas
D. solubility in water

A3 Topic: Carbohydrates CH5-LO2

An aldohexose has how many chiral carbons in its open-chain form?

A. 4
B. 3
C. 5
D. 6

A4 Topic: Carbohydrates CH5-LO2

A sugar is assigned D or L by the –OH on:

A. the chiral carbon furthest from the carbonyl group
B. C1
C. C2
D. the anomeric carbon

A5 Topic: Carbohydrates CH5-LO3

Anomers differ only at:

A. the anomeric carbon
B. C4
C. the CH_2OH group
D. every chiral carbon

A6 Topic: Carbohydrates CH5-LO3

At equilibrium in water, D-glucose is mostly:

A. β-D-glucose (about 64%)
B. α-D-glucose (about 64%)
C. open-chain glucose
D. fructose

A7 Topic: Carbohydrates CH5-LO4

A reducing sugar is one that:

A. reduces Benedict's or Tollens' reagent
B. is reduced to a sugar alcohol
C. cannot be hydrolysed
D. has no –OH groups

A8 Topic: Carbohydrates CH5-LO4

Sorbitol is formed from glucose by:

A. reduction
B. oxidation
C. hydrolysis
D. glycoside formation

A9 Topic: Carbohydrates CH5-LO5

Maltose is hydrolysed to:

A. two glucose
B. glucose + fructose
C. glucose + galactose
D. two fructose

A10 Topic: Carbohydrates CH5-LO5

The glycosidic linkage in lactose is:

A. β-1,4
B. α-1,4
C. α-1,β-2
D. α-1,6

A11 Topic: Carbohydrates CH5-LO5

Glycogen differs from amylopectin in being:

A. more highly branched
B. unbranched
C. made of fructose
D. linked β-1,4

A12 Topic: Carbohydrates CH5-LO5

Which statement about cellulose is correct?

A. Its glucose units are joined by β-1,4 links
B. It is a reducing sugar
C. It gives a blue-black colour with iodine
D. Humans digest it with amylase

Part B · Structured questions

Write your answers on paper, then self-mark with the worked solution.

Question B1 Topic: Carbohydrates CH5-LO1 6 marks, C3

(a) Define a chiral carbon and explain what enantiomers are. (2 marks)

(b) State whether each compound has a chiral carbon, and identify it if so: (3 marks)

(i) $CH_3CH(OH)COOH$ (lactic acid)

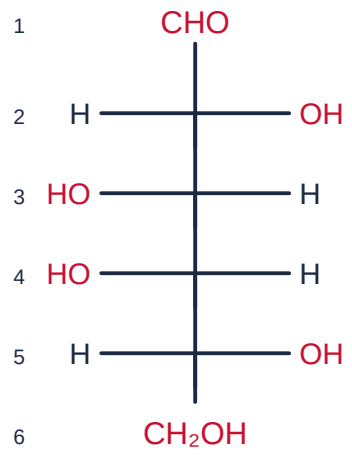
(ii) $CH_3CH(OH)CH_2CH_3$

(iii) $HOCH_2CH(OH)CH_2OH$ (glycerol).

(c) Ibuprofen is sold as a racemic mixture. Explain why a racemic mixture does not rotate plane-polarised light. (1 mark)

Question B2 Topic: Carbohydrates CH5-LO2 8 marks, C3

The Fischer projection of a monosaccharide is shown.



(a) Classify it by its carbonyl group and by the number of carbons. (2 marks)

(b) How many chiral carbons does it have, and what is the maximum number of stereoisomers with this skeleton? (2 marks)

(c) Is it a D or an L sugar? Explain. (2 marks)

(d) Name the sugar, and state how it differs from D-glucose. (2 marks)

Question B3 Topic: Carbohydrates CH5-LO3 8 marks, C3

(a) Explain how D-glucose forms a six-membered ring in water, and name the functional group formed. (2 marks)

(b) Draw the Haworth projections of α-D-glucose and β-D-glucose, and label the anomeric carbon. (4 marks)

(c) A freshly made solution of pure α-D-glucose has a specific rotation of +112°, which changes to +52.7°. Explain. (2 marks)

Question B4 Topic: Carbohydrates CH5-LO4 8 marks, C4

Four solutions are warmed with Benedict's reagent: glucose, fructose, sucrose and methyl α-D-glucoside.

(a) Predict the result for each and give a reason. (4 marks)

(b) The sucrose solution is warmed with dilute HCl, neutralised, and tested again. It now turns brick-red. Explain, naming the products. (2 marks)

(c) Name the product of reducing glucose with H_2/Ni, and give one pharmaceutical use of it. (2 marks)

Question B5 Topic: Carbohydrates CH5-LO5 8 marks, C4

(a) Compare maltose and lactose in terms of their monosaccharide units and glycosidic linkage. (2 marks)

(b) A patient gets bloating and diarrhoea after drinking milk. Name the condition and explain its cause. (2 marks)

(c) Compare amylose, amylopectin and cellulose in terms of linkage and branching. (3 marks)

(d) Microcrystalline cellulose is a common tablet filler. Explain why it passes through the gut undigested. (1 mark)

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

Adapted from *The Basics of GOB Chemistry* (Ball, Hill & Scott) chapter 16, and *Fundamentals of General Organic and Biological Chemistry* 20.3–20.7, LibreTexts, CC BY-NC-SA 3.0. Chirality and drug questions, sugar alcohols and pharmacy scenarios are original material by Dr. NAM. Shared under CC BY-NC-SA for non-commercial teaching.

PHD125 Pharmaceutical Organic Chemistry · Carbohydrates

Tutorial: exam-style practice

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

Part A: 12 multiple-choice questions (1 mark each). Part B: 5 structured questions (38 marks). Same format as the final exam.

Answer key

Part A · Multiple choice

A1 Topic: Carbohydrates CH5-LO1

Carbohydrates are best defined as:

A. polyhydroxy aldehydes or ketones, or compounds that hydrolyse to them

B. compounds of carbon and water molecules

C. esters of glycerol

D. polymers of amino acids

Ball 16.1.

A2 Topic: Carbohydrates CH5-LO1

Enantiomers differ in their:

A. direction of rotation of plane-polarised light

B. boiling points

C. molecular formulas

D. solubility in water

All other physical properties are the same.

A3 Topic: Carbohydrates CH5-LO2

An aldohexose has how many chiral carbons in its open-chain form?

A. 4

B. 3

C. 5

D. 6

C2–C5; 2^4 = 16 stereoisomers.

A4 Topic: Carbohydrates CH5-LO2

A sugar is assigned D or L by the –OH on:

A. the chiral carbon furthest from the carbonyl group

B. C1

C. C2

D. the anomeric carbon

The penultimate carbon (Ball 16.2).

A5 Topic: Carbohydrates CH5-LO3

Anomers differ only at:

A. the anomeric carbon

B. C4

C. the CH_2OH group

D. every chiral carbon

C1 for aldoses, C2 for fructose.

A6 Topic: Carbohydrates CH5-LO3

At equilibrium in water, D-glucose is mostly:

A. β-D-glucose (about 64%)

B. α-D-glucose (about 64%)

C. open-chain glucose

D. fructose

About 36% α, 64% β, under 0.02% open chain.

A7 Topic: Carbohydrates CH5-LO4

A reducing sugar is one that:

A. reduces Benedict's or Tollens' reagent

B. is reduced to a sugar alcohol

C. cannot be hydrolysed

D. has no –OH groups

Cu^{2+} → Cu_2O.

A8 Topic: Carbohydrates CH5-LO4

Sorbitol is formed from glucose by:

A. reduction

B. oxidation

C. hydrolysis

D. glycoside formation

C=O → CH–OH.

A9 Topic: Carbohydrates CH5-LO5

Maltose is hydrolysed to:

A. two glucose

B. glucose + fructose

C. glucose + galactose

D. two fructose

Maltase or dilute acid.

A10 Topic: Carbohydrates CH5-LO5

The glycosidic linkage in lactose is:

A. β-1,4

B. α-1,4

C. α-1,β-2

D. α-1,6

Galactose C1 (β) to glucose C4.

A11 Topic: Carbohydrates CH5-LO5

Glycogen differs from amylopectin in being:

A. more highly branched

B. unbranched

C. made of fructose

D. linked β-1,4

Branches every 8–12 units instead of 25–30.

A12 Topic: Carbohydrates CH5-LO5

Which statement about cellulose is correct?

A. Its glucose units are joined by β-1,4 links

B. It is a reducing sugar

C. It gives a blue-black colour with iodine

D. Humans digest it with amylase

FGOB 20.7.

Part B · Structured questions

Question B1 Topic: Carbohydrates CH5-LO1 6 marks, C3

(a) Define a chiral carbon and explain what enantiomers are. (2 marks)

(b) State whether each compound has a chiral carbon, and identify it if so: (3 marks)

- **(i)** $CH_3CH(OH)COOH$ (lactic acid)
- **(ii)** $CH_3CH(OH)CH_2CH_3$
- **(iii)** $HOCH_2CH(OH)CH_2OH$ (glycerol).

(c) Ibuprofen is sold as a racemic mixture. Explain why a racemic mixture does not rotate plane-polarised light. (1 mark)

(a) A chiral carbon is bonded to four different groups. Enantiomers are a pair of stereoisomers that are non-superimposable mirror images of each other.

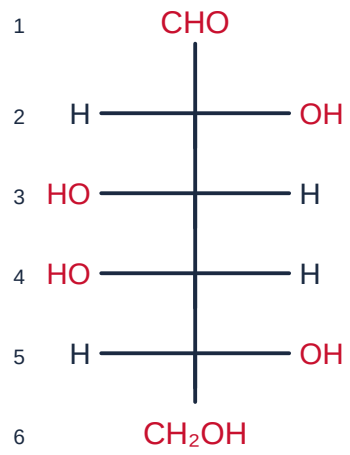
(b)
- **(i)** Yes: C2 (H, OH, CH_3, COOH)
- **(ii)** Yes: C2 (H, OH, CH_3, CH_2CH_3)
- **(iii)** No: C2 carries two identical CH_2OH groups.

(c) It contains equal amounts of the two enantiomers, which rotate light by equal amounts in opposite directions, so the rotations cancel.

Step	Marking criteria	Marks
1	Chiral carbon: four different groups	1
2	Enantiomers: non-superimposable mirror images	1
3	Lactic acid: yes, C2	1
4	2-Butanol: yes, C2	1
5	Glycerol: no (two identical CH_2OH)	1
6	Equal and opposite rotations cancel	1
	Total	**6**

Question B2 Topic: Carbohydrates CH5-LO2 8 marks, C3

The Fischer projection of a monosaccharide is shown.



(a) Classify it by its carbonyl group and by the number of carbons. (2 marks)

(b) How many chiral carbons does it have, and what is the maximum number of stereoisomers with this skeleton? (2 marks)

(c) Is it a D or an L sugar? Explain. (2 marks)

(d) Name the sugar, and state how it differs from D-glucose. (2 marks)

(a) Aldose (aldehyde at C1); hexose (6 carbons): an aldohexose.

(b) Four chiral carbons (C2–C5); 2^4 = 16 stereoisomers.

(c) D: the –OH on C5, the chiral carbon furthest from the carbonyl group, is on the right.

(d) D-galactose. It differs from D-glucose only at C4, where its –OH is on the left.

Step	Marking criteria	Marks
1	Aldose	1
2	Hexose	1
3	4 chiral carbons	1
4	16 stereoisomers	1
5	D sugar	1
6	Reason: C5–OH on the right	1
7	D-galactose	1
8	Differs at C4 (OH on the left)	1

	Total	8

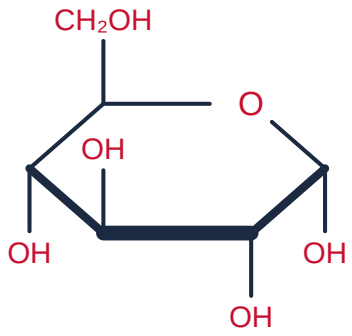
Question B3 Topic: Carbohydrates CH5-LO3 8 marks, C3

(a) Explain how D-glucose forms a six-membered ring in water, and name the functional group formed. (2 marks)

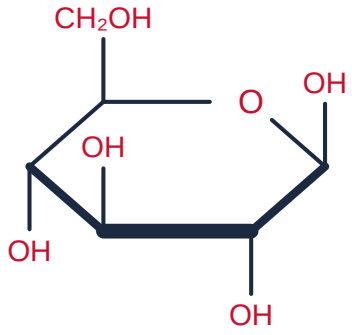
(b) Draw the Haworth projections of α-D-glucose and β-D-glucose, and label the anomeric carbon. (4 marks)

(c) A freshly made solution of pure α-D-glucose has a specific rotation of +112°, which changes to +52.7°. Explain. (2 marks)

(a) The –OH on C5 adds to the C1 aldehyde (C=O) in the same molecule, giving a ring of five carbons and one oxygen; the new group is a hemiacetal.



α-D-glucose



β-D-glucose

(b) In both: ring O at the back right; C2–OH down, C3–OH up, C4–OH down, CH_2OH up on C5. α: C1–OH down; β: C1–OH up. C1 is the anomeric carbon.

(c) In water the ring opens and recloses (mutarotation), forming an equilibrium mixture of about 36% α and 64% β (plus a trace of open chain), whose rotation is +52.7°.

Step	Marking criteria	Marks
1	C5–OH adds to the C1 aldehyde	1
2	Hemiacetal	1

3	Correct ring with O and CH_2OH up	1
4	C2, C3, C4 –OH positions correct	1
5	α: C1–OH down; β: C1–OH up	1
6	Anomeric carbon (C1) labelled	1
7	Ring opens and recloses: mutarotation	1
8	Equilibrium ≈ 36% α, 64% β	1
	Total	**8**

Question B4 Topic: Carbohydrates CH5-LO4 8 marks, C4

Four solutions are warmed with Benedict's reagent: glucose, fructose, sucrose and methyl α-D-glucoside.

(a) Predict the result for each and give a reason. (4 marks)

(b) The sucrose solution is warmed with dilute HCl, neutralised, and tested again. It now turns brick-red. Explain, naming the products. (2 marks)

(c) Name the product of reducing glucose with H_2/Ni, and give one pharmaceutical use of it. (2 marks)

(a) Glucose: brick-red; its ring opens to an aldehyde (reducing sugar). Fructose: brick-red; in the basic reagent it rearranges to an aldose form. Sucrose: stays blue; both anomeric carbons are in the glycosidic bond, so it cannot open (non-reducing). Methyl α-D-glucoside: stays blue; the anomeric carbon is locked as a glycoside.

(b) Acid hydrolyses sucrose to glucose and fructose (invert sugar), which are reducing sugars.

(c) Sorbitol (glucitol); a sweetener in sugar-free syrups and lozenges (or a laxative).

Step	Marking criteria	Marks
1	Glucose: brick-red, reducing (opens to aldehyde)	1
2	Fructose: brick-red (rearranges in base)	1
3	Sucrose: blue, both anomeric C used	1
4	Glycoside: blue, anomeric C locked	1
5	Hydrolysis of sucrose	1
6	Glucose + fructose (invert sugar) are reducing	1
7	Sorbitol	1
8	A valid use	1
	Total	**8**

Question B5 Topic: Carbohydrates CH5-LO5 8 marks, C4

(a) Compare maltose and lactose in terms of their monosaccharide units and glycosidic linkage. (2 marks)

(b) A patient gets bloating and diarrhoea after drinking milk. Name the condition and explain its cause. (2 marks)

(c) Compare amylose, amylopectin and cellulose in terms of linkage and branching. (3 marks)

(d) Microcrystalline cellulose is a common tablet filler. Explain why it passes through the gut undigested. (1 mark)

(a) Maltose: two glucose units joined α-1,4. Lactose: galactose + glucose joined β-1,4.

(b) Lactose intolerance: too little of the enzyme lactase, so lactose is not hydrolysed; it draws water into the colon and bacteria ferment it, producing acid and gas.

(c) Amylose: α-1,4, unbranched (helix). Amylopectin: α-1,4 with α-1,6 branches every 25–30 units. Cellulose: β-1,4, unbranched straight chains that hydrogen bond into fibres.

(d) Humans have no enzyme that hydrolyses β-1,4-glycosidic bonds.

Step	Marking criteria	Marks
1	Maltose: glucose + glucose, α-1,4	1
2	Lactose: galactose + glucose, β-1,4	1
3	Lactose intolerance	1
4	Too little lactase (cause and effect)	1
5	Amylose: α-1,4 unbranched	1
6	Amylopectin: α-1,4 + α-1,6 branches	1
7	Cellulose: β-1,4 unbranched	1
8	No human enzyme for β-1,4	1
	Total	**8**

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

Adapted from *The Basics of GOB Chemistry* (Ball, Hill & Scott) chapter 16, and *Fundamentals of General Organic and Biological Chemistry* 20.3–20.7, LibreTexts, CC BY-NC-SA 3.0. Chirality and drug questions, sugar alcohols and pharmacy scenarios are original material by Dr. NAM. Shared under CC BY-NC-SA for non-commercial teaching.